



Feedback form for the course

Please do fill this in before the end of term

We do take comments seriously, and they help to make the course better over time

https://cambridge.eu.qualtrics.com/jfe/form/SV_3z0oG7tq3O9xtpc

Equilibrium figures

From last lecture

Lagrangian in a self-gravitating planet:

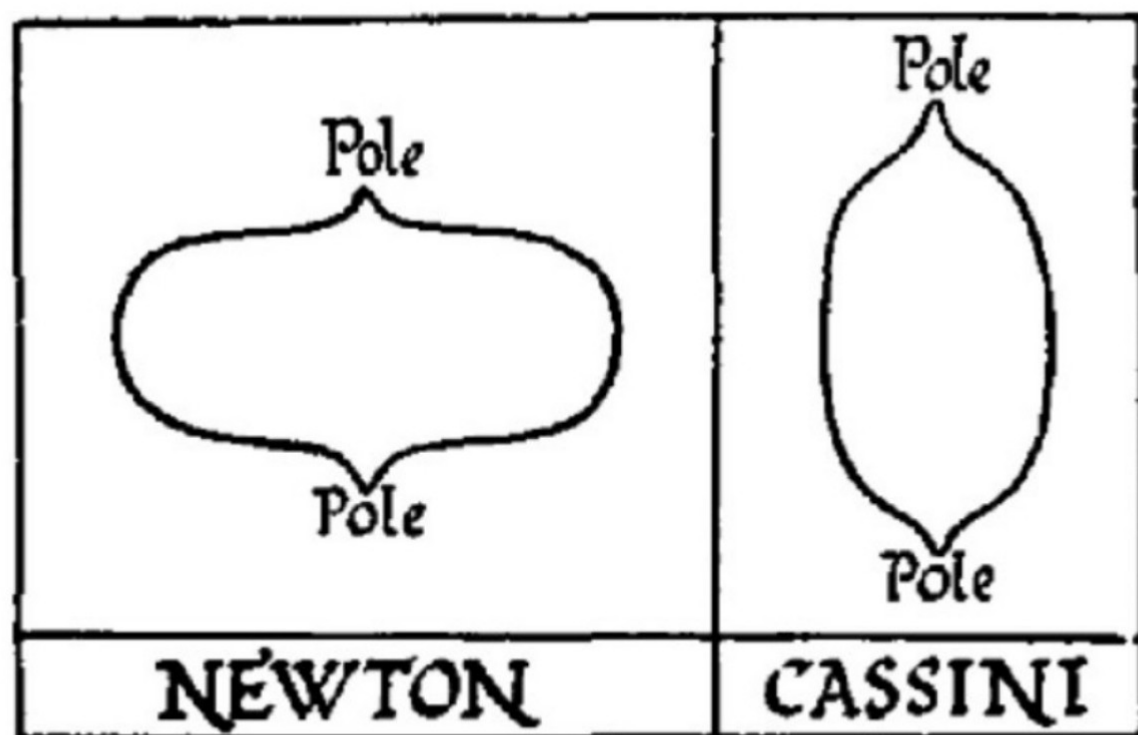
$$\mathcal{L}(\mathbf{x}, t, \varphi, \mathbf{v}, \mathbf{F}) = \frac{1}{2}\rho(\mathbf{x})||\mathbf{v}||^2 - W(\mathbf{x}, t, \mathbf{F}) - \frac{1}{2}\rho(\mathbf{x})\zeta(\mathbf{x}, t),$$

Decomposition of the motion relative to an approximately rotating frame:

$$\varphi_i(\mathbf{x}, t) = \Phi_i(t) + R_{ij}(t)\bar{\varphi}_j(\mathbf{x}, t),$$

Equilibrium equations:

$$\rho\epsilon_{ijk}\epsilon_{klm}\Omega_j\Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho\gamma_i^0 = 0,$$



“Vouz avez confirmé dans les lieux pleins d’ennui
Ce que Newton connut sans sortir de chez lui.”

Solutions of the equilibrium equations

Simplified notations

Notation from last lecture:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial T_{ij}^0}{\partial x_j} - \rho \gamma_i^0 = 0,$$

To simplify things we now write:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0,$$

At equilibrium referential and spatial quantities coincide

The stress tensor is symmetric and so has six independent components

Under-determined system

Suppose the planet's shape and density are known

We then have **three** equations for the **six** components of stress:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m - \frac{\partial \sigma_{ij}}{\partial x_j} - \rho g_i = 0,$$

This is an under-determined system and so we **cannot expect unique solutions**

More formally, there exists **non-trivial divergence-free symmetric second order tensor fields** that satisfy the correct boundary conditions

Compare with the existence of divergence-free vector fields

Hydrostatic equilibrium

Stress in a viscous fluid

The (Cauchy) stress in a compressible Newtonian fluid:

$$\sigma_{ij} = -p \delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} \frac{\partial v_k}{\partial x_k} \delta_{ij} \right)$$

At equilibrium, the stress is **hydrostatic**:

$$\sigma_{ij} = -p \delta_{ij}$$

The same result holds in more general viscous fluids

Scaling estimate for deviatoric stress associated with convection:

$$\eta V/L \sim 10^7 \text{ Pa}$$

Pressures in the Earth's interior are of order 10^{10} Pa – deviation from hydrostatic state in Earth-like planets represents only a **small perturbation**

Hydrostatic equilibrium equations

Given the hydrostatic approximation, we have:

$$\rho \epsilon_{ijk} \epsilon_{klm} \Omega_j \Omega_l x_m + \frac{\partial p}{\partial x_i} - \rho g_i = 0.$$

This is now an **over-determined** set of equations – three equations for one unknown

Helpful to switch to a vector-calculus notation:

$$\nabla p + \rho \nabla \gamma = 0,$$

Here we have introduced the **gravity-potential**, this being the **sum of the gravitational and centrifugal potentials**

Consequences of the equations

Hydrostatic equilibrium equation:

$$\nabla p + \rho \nabla \gamma = 0,$$

Gradients of pressure and gravity potential are parallel, and so a level surface of one field is a level surface of the other

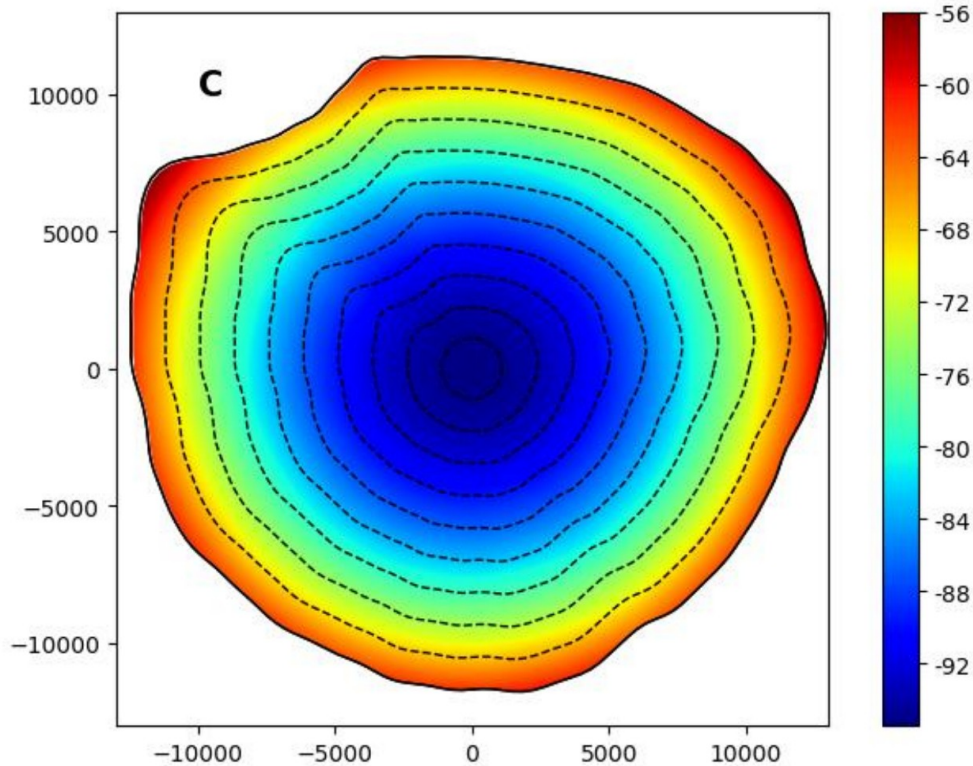
Ignoring the atmosphere, the pressure vanishes at the planet's surface, and hence this must be a level surface of the gravity potential

Taking the curl of the equation, we find:

$$\nabla \rho \times \nabla \gamma = 0,$$

This shows that level surfaces of density are also coincident with the other two fields

Gravitational potential for Phobos



Equatorial slice through the gravitational potential for Phobos calculated assuming a constant interior density

The potential field is non constant on the moon's surface, and hence this cannot be a hydrostatic equilibrium state

In general, the hydrostatic equilibrium condition imposes very strong conditions on a planet's shape, density, and rotation

Non-rotating planets

In the non-rotating case, suppose the planet is spherically symmetric

Equilibrium equation reduces to:

$$\nabla p + \rho \frac{d\phi}{dr} \hat{\mathbf{r}} = \mathbf{0},$$

There is just one non-trivial equation here, and so we can find an equilibrium solution

It can be shown that this is the **only** solution in the non-rotating case

Slowly rotating planets

By definition, **slowly rotating** means that the centrifugal potential is small relative to the gravitational potential:

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim \frac{3\Omega^2}{8\pi G\bar{\rho}}$$

For the Earth, we find:

$$\frac{\text{centrifugal potential}}{\text{gravitational potential}} \sim 2 \times 10^{-3},$$

In such cases, we can apply first-order perturbation theory

Higher-order theories exist and are sometimes necessary

Perturbation expansions

Expand about the non-rotating solution:

$$\rho = \rho_0 + s\rho_1 + \cdots,$$

$$p = p_0 + s p_1 + \cdots,$$

$$\phi = \phi_0 + s \phi_1 + \cdots,$$

$$\psi = s \psi_1,$$

Similarly, the free-surface radius is expanded as:

$$r(\theta, \varphi) = b + s h_1(\theta, \varphi) + \cdots,$$

Topography on internal boundaries expanded similarly, but we neglect this detail for simplicity in what follows

First-order equilibrium equations

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

Exact boundary condition on the perturbed planetary surface:

$$p[b + s h_1(\theta, \varphi) + \dots, \theta, \varphi] = 0.$$

Expanded to first-order, this reduces to:

$$p_1 + \frac{dp_0}{dr} h_1 = 0$$

$$p_1 - \rho_0 g_0 h_1 = 0.$$

Analysis of the equations

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

Take cross-product with radial unit vector:

$$\hat{\mathbf{r}} \times \nabla p_1 + \hat{\mathbf{r}} \times \rho_0 \nabla \gamma_1 = \mathbf{0}.$$

$$\hat{\mathbf{r}} \times \nabla (p_1 + \rho_0 \gamma_1) = \mathbf{0}.$$

From this, we conclude that:

$$p_1 = -\rho_0 \gamma_1$$

Analysis of the equations

First-order perturbation of the equilibrium equation:

$$\nabla p_1 + \rho_0 \nabla \gamma_1 + \rho_1 g_0 \hat{\mathbf{r}} = \mathbf{0},$$

Taking the curl of this equation, we arrive at:

$$\hat{\mathbf{r}} \times \nabla \left(\frac{d\rho_0}{dr} \gamma_1 - g_0 \rho_1 \right) = \mathbf{0},$$

And hence, we see that:

$$\rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1$$

Analysis of the equations

We have now shown that the following identities hold for the first-order perturbations:

$$p_1 = -\rho_0 \gamma_1, \quad h_1 = -\frac{\gamma_1}{g_0}, \quad \rho_1 = \frac{1}{g_0} \frac{d\rho_0}{dr} \gamma_1.$$

All other terms can be expressed through the gravity potential perturbation

Linearised Poisson equation

Linearisation of Poisson's equation is trivial:

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \rho_1 & r < b \\ 0 & r > b \end{cases}$$

Linearisation of continuity conditions requires more work:

$$[\phi_1]_{-}^{+} = 0, \quad [\hat{\mathbf{r}} \cdot \nabla \phi_1 + 4\pi G \rho_0 h_1]_{-}^{+} = 0$$

Derivation done within the final example sheet

Modified Poisson equation

Expressing other perturbations in terms of the gravity potential:

$$\nabla^2 \phi_1 = \begin{cases} 4\pi G \frac{1}{g_0} \frac{d\rho_0}{dr} (\phi_1 + \psi_1) & r < b \\ 0 & r > b \end{cases}$$

Expressing other perturbations in terms of the gravity potential:

$$[\phi_1]_{-}^{+} = 0, \quad \left[\hat{\mathbf{r}} \cdot \nabla \phi_1 - 4\pi G \rho_0 \frac{(\phi_1 + \psi_1)}{g_0} \right]_{-}^{+} = 0.$$

Linear boundary value problem for the perturbed gravitational potential, with the centrifugal potential providing the “force terms”

Theory applies also to equilibrium tides in gaseous planets

Ellipticity function

Separation of variables shows that:

$$\phi_1 + \psi_1 = \sqrt{\frac{4\pi}{5}} \frac{2}{3} r g_0 \epsilon Y_{20}.$$

which serves to define the **ellipticity function**

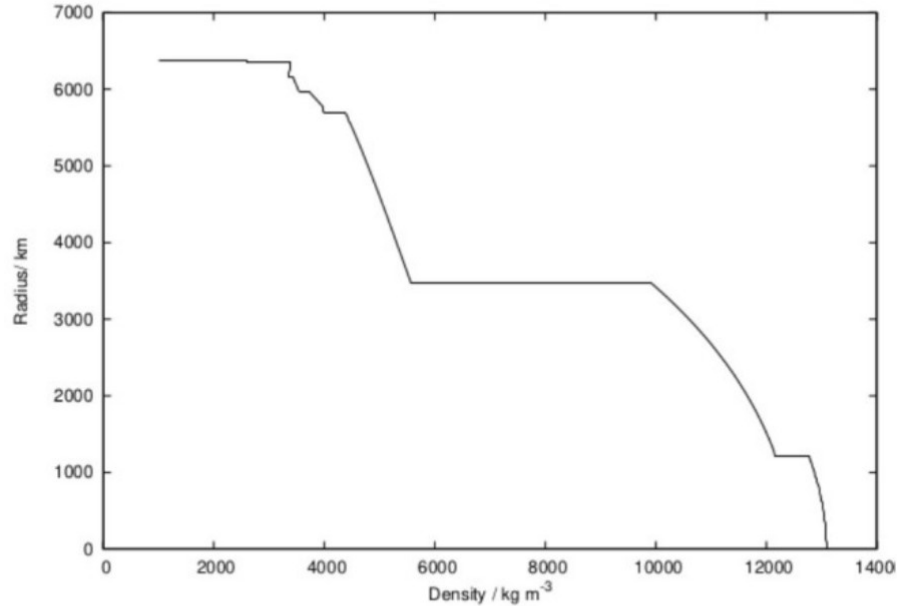
This function has the property:

$$\epsilon(b) = \frac{\text{equatorial radius} - \text{polar radius}}{\text{mean radius}}$$

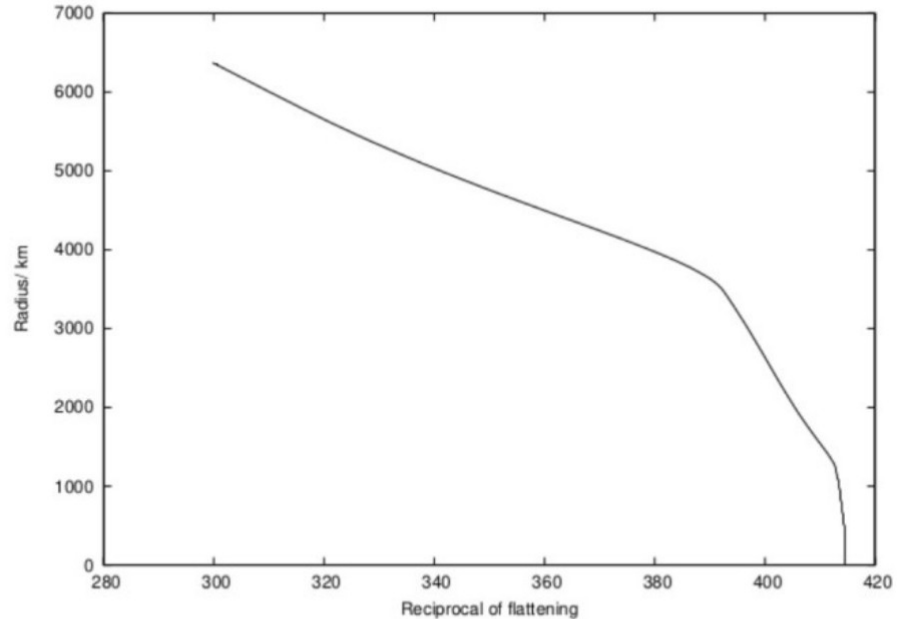
at the planets mean surface (and at internal ones too)

Surface ellipticity of the Earth is around 1/300

Numerical calculation for the Earth



Radial density profile obtained from analysis of the Earth's free oscillations – final two lectures



Corresponding value of ellipticity (or flattening) plotted here as a reciprocal

What is a planet?



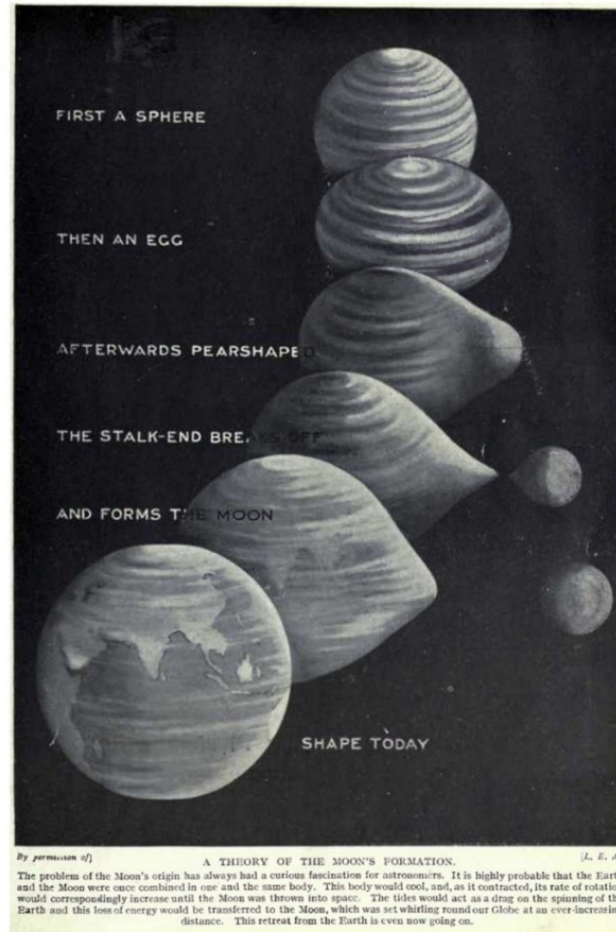
A planet



Not a planet



Haumea – A rapidly rotating (4hr period) Dwarf planet with a **triaxial** equilibrium figure



Fission theory for moon formation due to George Darwin. This idea is generally now dismissed, but it does have a real physical basis.